

Use of ML and XAI as decision–support for Business Intelligence analysis.

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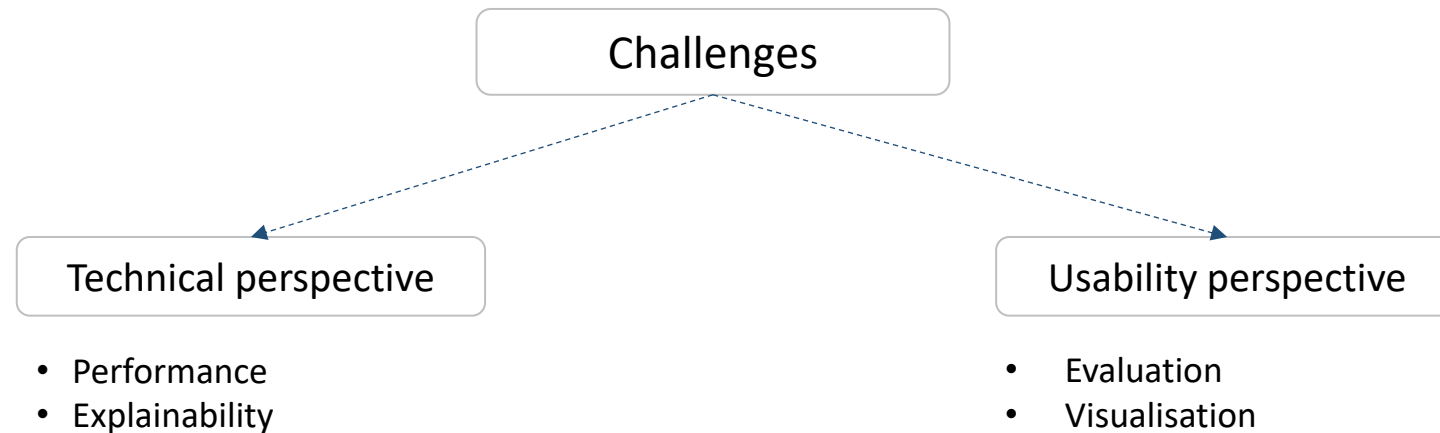
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Decision suport challenges

Align business objectives with technical aspects of ML model training
Universality or specialization?

Add context for interpretation of XAI results to enhance usefulness of predictions



Inefficient model → **wrong conclusions**

Unclear results → **no conclusions**

Usability of ML in Business Intelligence

Motivations for investment in Industry 4.0 - Company **profit**

Revenue

Cost optimisation

Time optimisation

Quality optimisation

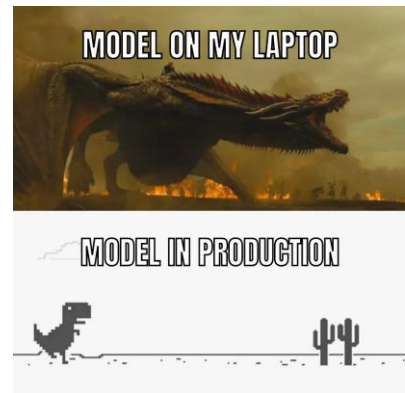
Tasks of ML as Decision Support

Mimic current abilities

Interpretable and consistent with domain knowledge

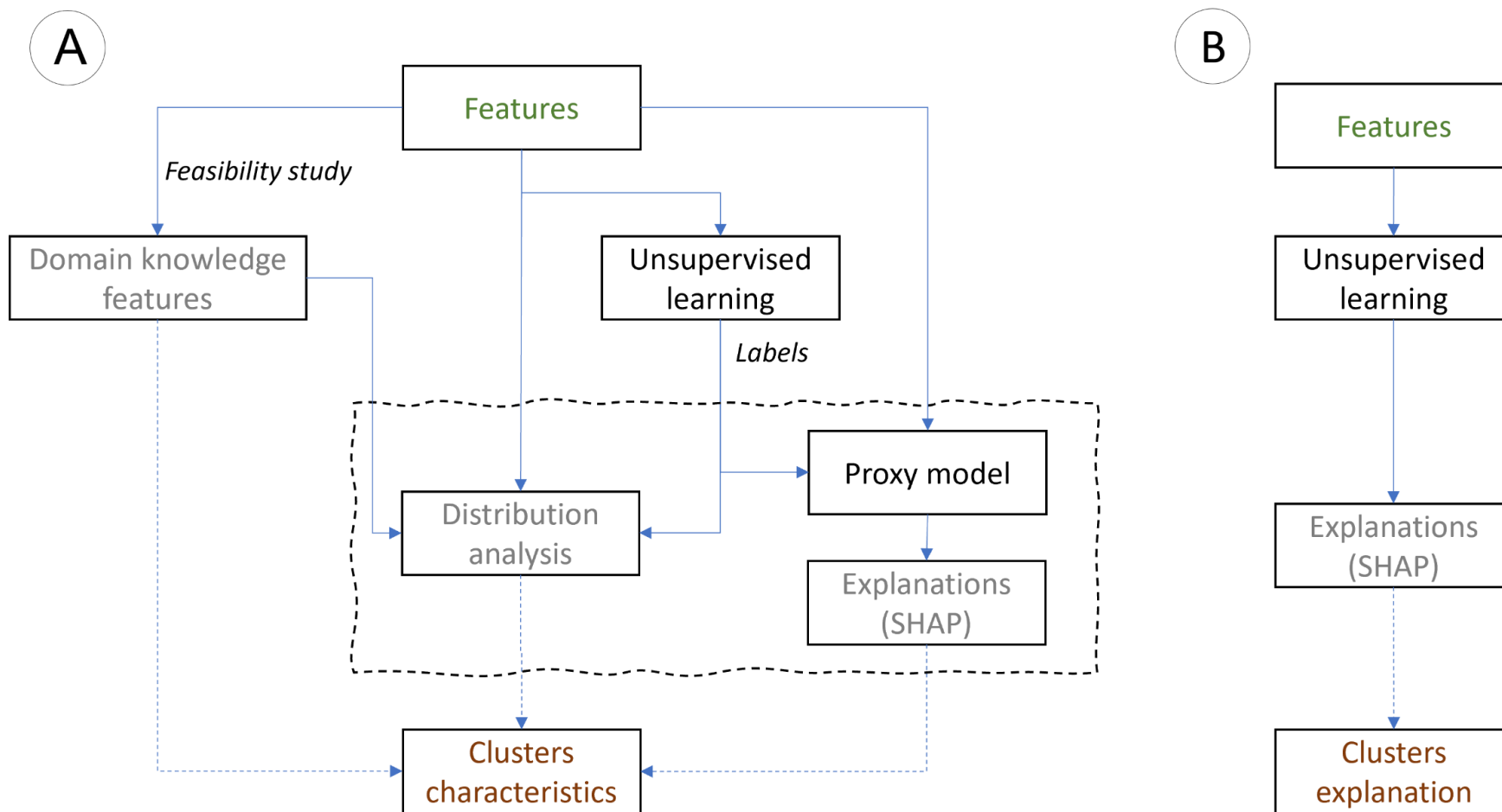
Enhance current abilities

Need of new metrics, and context for new results interpretation

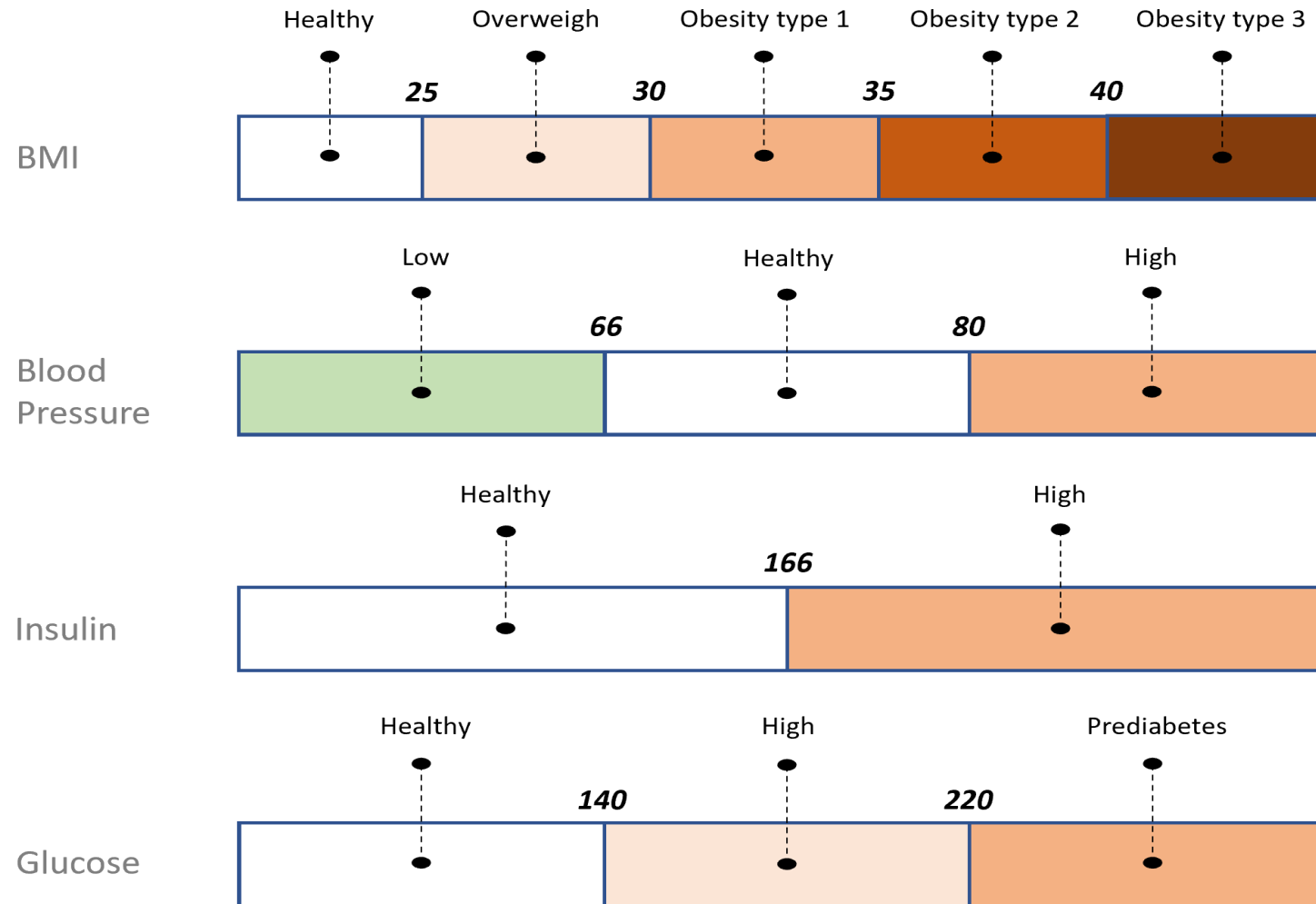


„Improving understandability of explanations with a usage of expert knowledge – unsupervised learning”

Maciej Szelążek, Szymon Bobek, Grzegorz J. Nalepa



Feasibility study of domain knowledge



Multiple information sources for describing clusters

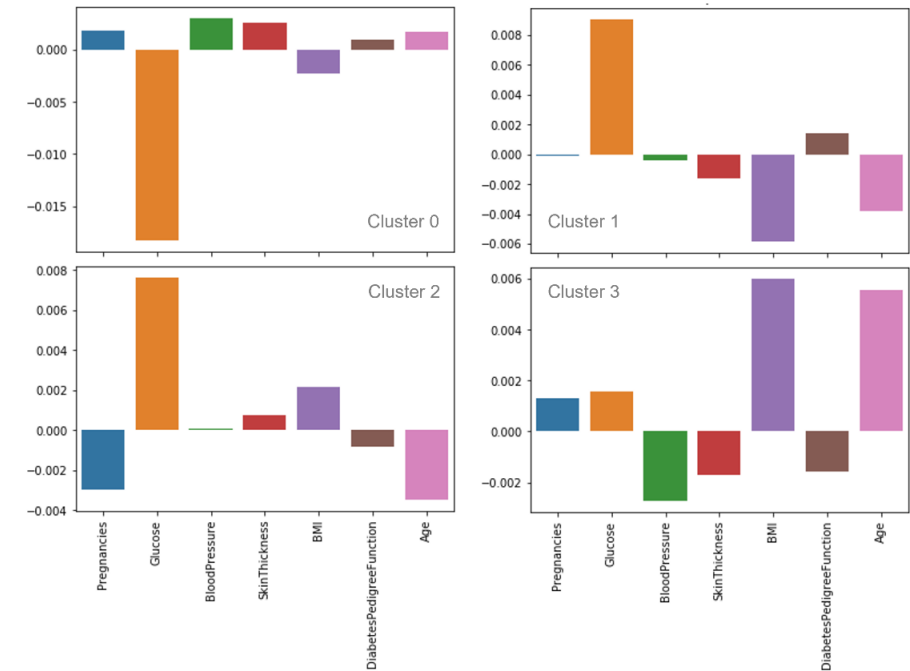
Original features characteristics

Cluster	Pregnancies	Glucose	BloodPressure	SkinThickness	BMI	DiabetesPedigreeFunction	Age
0	5,0	168,7	76,6	25,1	35,1	0,55	38,7
1	3,5	121,7	73,7	33,6	34,9	0,50	31,9
2	4,6	121,9	75,9	2,3	30,7	0,42	37,7
3	2,9	90,8	66,2	21	29,8	0,44	28,2

Domain features characteristics

Cluster	BMI_obese3	BMI_obese2	BMI_obese1	BMI_overweight	Glucose_hi	BloodPressure_hi	BloodPressure_lo
0	32	33	57	20	153	54	24
1	34	53	58	40	16	51	46
2	12	18	42	42	16	40	25
3	12	37	55	64	0	18	111

SHAP characteristics

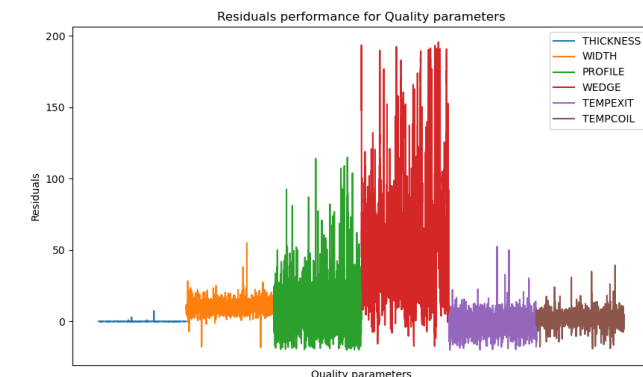


„Use of model explainability as actionable Decision–Support for data series analysis”

Maciej Szelązek, Szymon Bobek, Grzegorz J. Nalepa

Goal: Add context for interpretation of XAI results

- Contextualisation - Presentation of XAI in the context of quality indicators consistent with the current methodology of quality management in the organization
- Actionability
 - Automation of the procedure
 - Universal KPIs - The procedure allows for any dependent variable (quality metric), regardless of the input data
- Visualisation
 - Present the results in a way that can be understood by a diverse range of audiences



KPIs - Capability factors

Process capability – Cp : tolerance width divided by the total spread of process (6 Sigma).

Calculation of Process Capability (Cp) :

$$Cp = \frac{\text{Design Tolerance}}{6\sigma} = \frac{USL - LSL}{6\sigma}$$

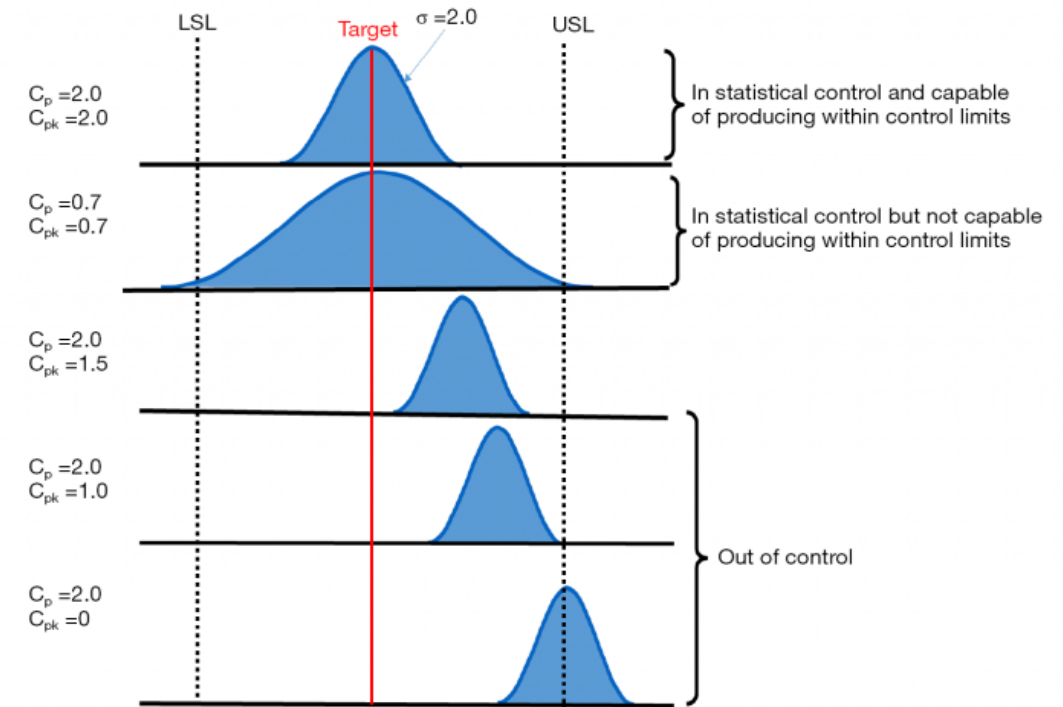
USL = Upper Specification Limit, LSL = Lower Specification Limit

Process Capability Index – Cpk: indicates shifting of the process, the minimum of Cpk upper and Cpk lower.

Calculation of Process Capability Index (Cpk) :

$$Cpk_U = \frac{USL - \bar{X}}{3\sigma} \text{ and } Cpk_L = \frac{\bar{X} - LSL}{3\sigma}$$

Cp	Defects amount
1	2700 ppm
1,33	63 ppm
1,67	0,57 ppm
2	0,002 ppm



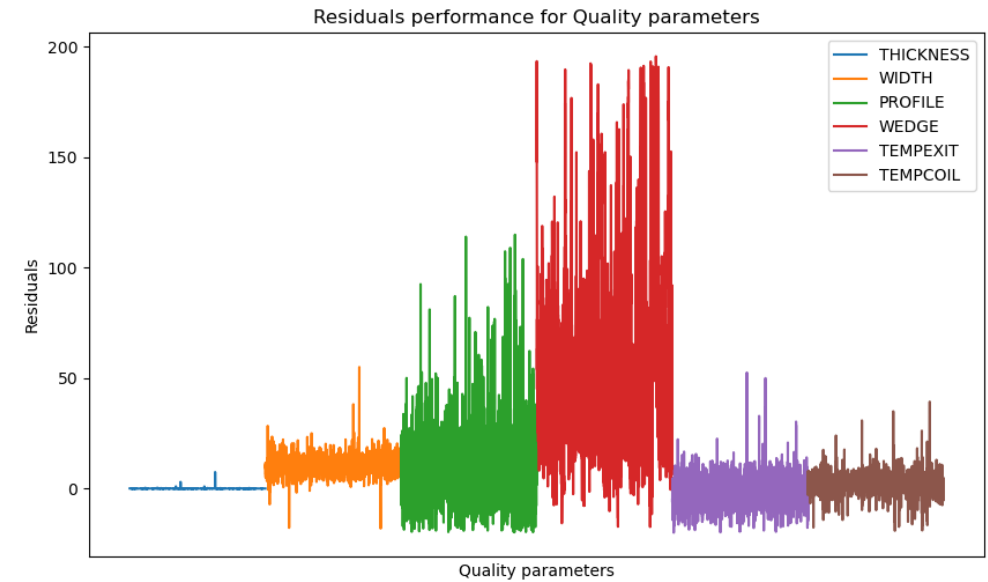
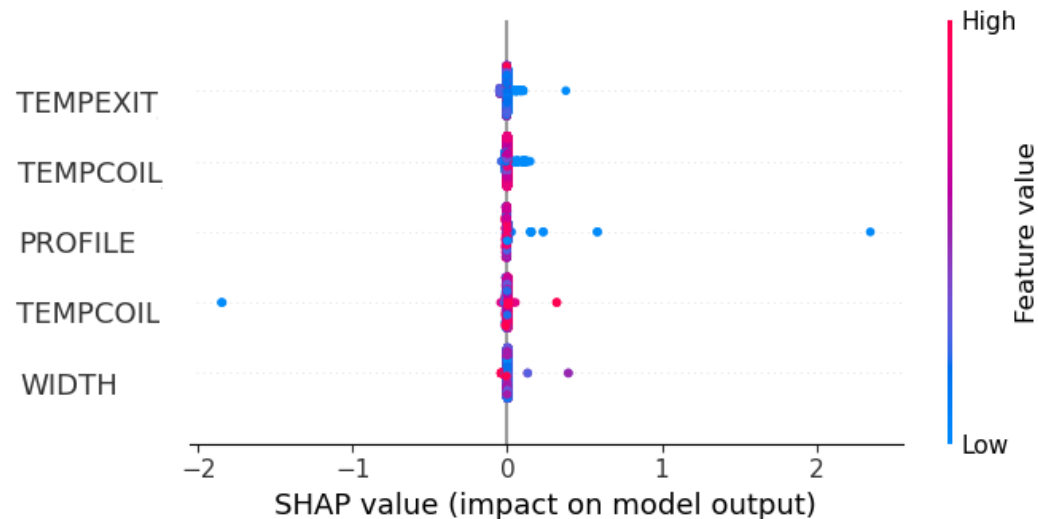
Issue – how to interpret XAI results

Lots of information based on analysis of KPI residuals:

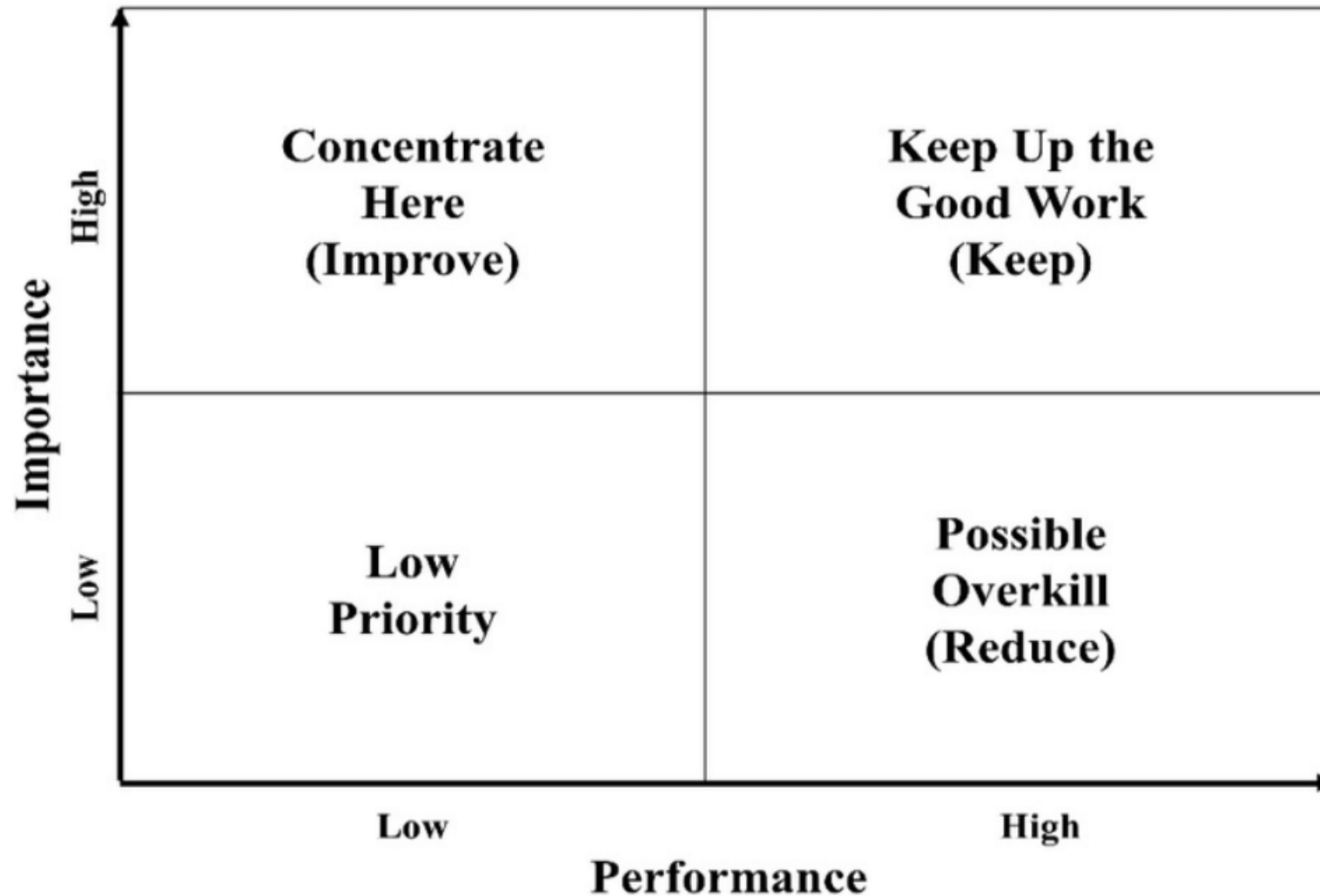
- Different levels of overruns – amplitude
- Trends above and below target – density
- Cases significantly different from natural variability - outliers

SHAP

- Similar XAI values for extremely different feature values
- No clear relationships between XAI values for different features



Importance Performance Analysis

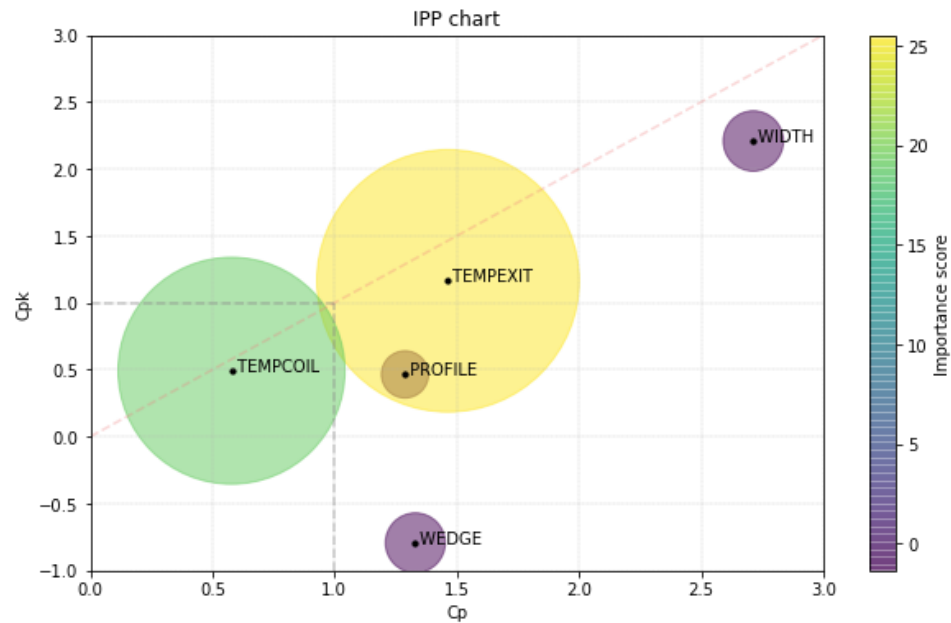


Results - IPP chart

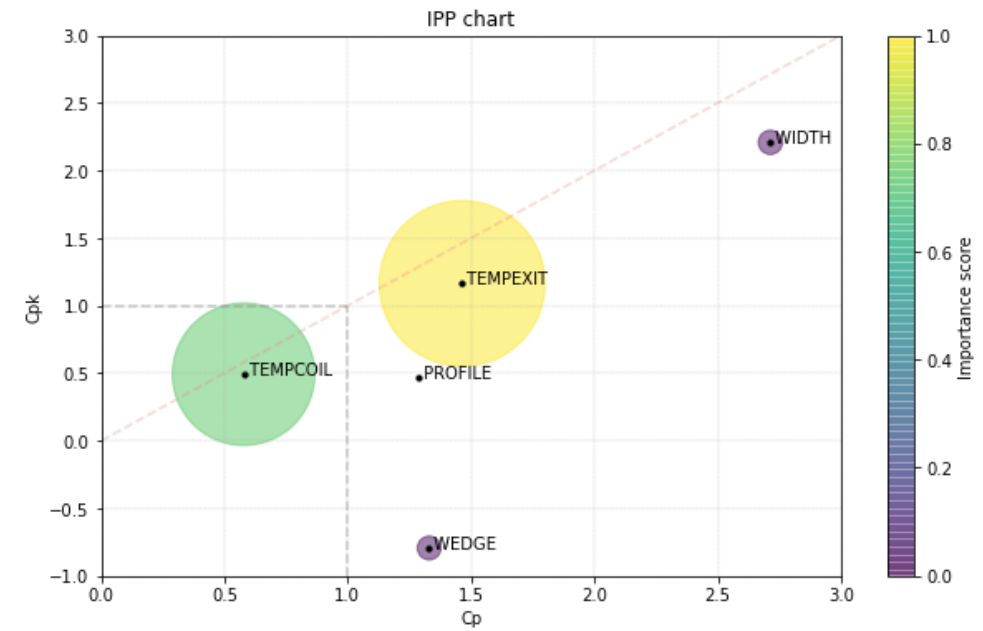
The closer to the **red line** the more centered the process

The further outside the gray lines the better.

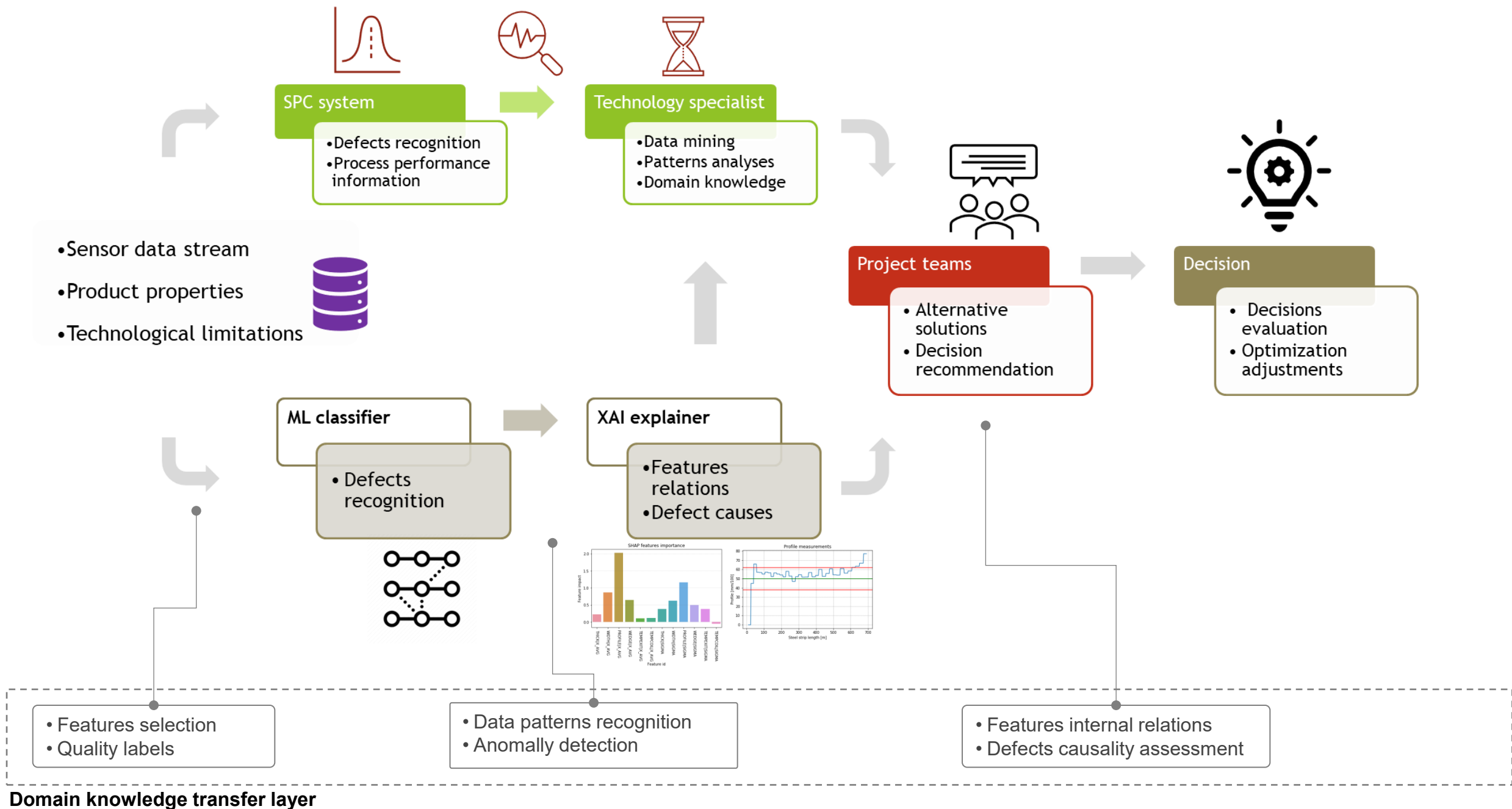
The table shows the sign of the XAI score, and the precise scale of differences between the features importances.



Feature	Cp	Cpk	residuals shap
WIDTH	2.71	2.21	-1.35
PROFILE	1.29	0.46	-0.83
WEDGE	1.33	-0.79	-1.33
TEMPEXIT	1.46	1.16	25.54
TEMPCOIL	0.58	0.49	19.10



Feature	Cp	Cpk	residuals shap
WIDTH	2.71	2.21	0.02
PROFILE	1.29	0.46	0.00
WEDGE	1.33	-0.79	0.02
TEMPEXIT	1.46	1.16	1.00
TEMPCOIL	0.58	0.49	0.74



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Thank you for your attention